Critique & Redesign Workshop

Workshop 1 — Module 7

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Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

1. The four-quadrant critique framework
2. Five canonical redesign case studies
3. The redesign workflow: identify $\rightarrow$ diagnose $\rightarrow$ fix $\rightarrow$ validate
4. Peer review protocol for visualization

Learning Outcome
You will be able to systematically critique any chart against four quality dimensions and execute a principled redesign using the techniques from Modules 1–6.

The Critique Framework

Four Quadrants of Critique

The Four-Quadrant Critique Framework

1. DATA INTEGRITY

- Is the data accurate?
- Are axes labelled correctly?
- Lie factor ≈ 1.0?
- No cherry-picking?

2. VISUAL ENCODING

- Best channel for each variable?
- Cleveland ranking respected?
- Appropriate chart type?
- Colour palette correct?

CRITIQUE RUBRIC

3. DESIGN CRAFT

- Typographic hierarchy clear?
- Data-ink ratio high?
- Whitespace adequate?
- Consistent style?

4. COMMUNICATION

- Clear title states the finding?
- Audience appropriate?
- Annotations guide reading?
- Actionable insight?

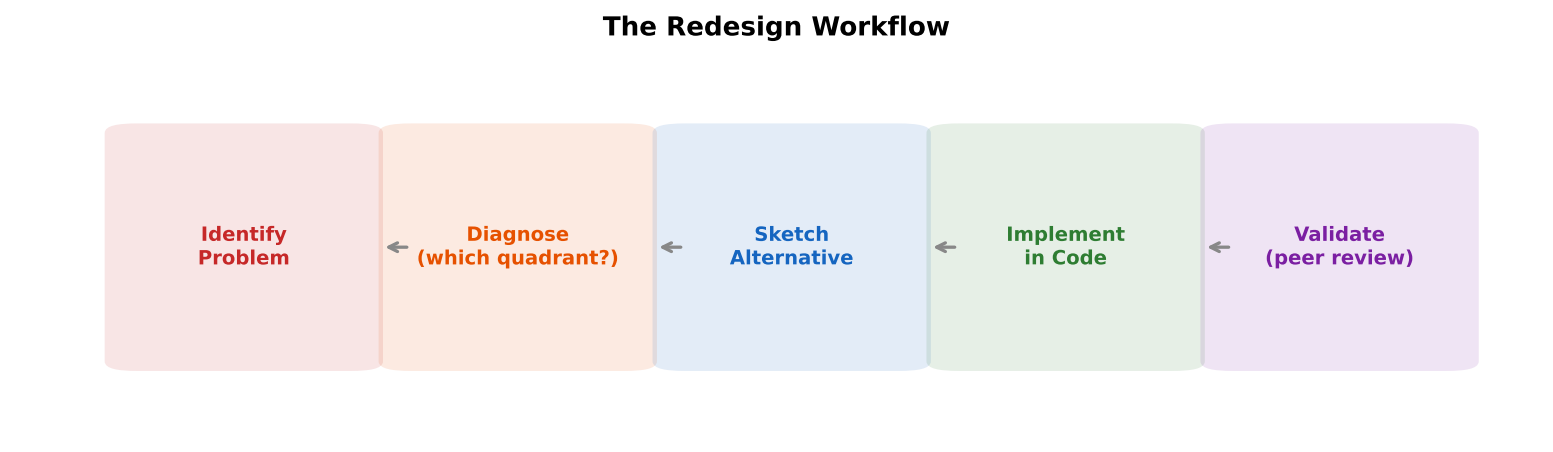
Using the Framework

Quadrant	Key Question	Module Ref.
Data Integrity	Does it honestly represent the data?	M01, M02
Visual Encoding	Is each variable on the best channel?	M03, M04
Design Craft	Is the typography/-colour/whitespace clean?	M04, M05
Communication	Does the audience get the message?	M05, M06

Pro-Tip
Critique the chart, not the author. Use the framework as an objective diagnostic tool: "The lie factor is 3.5 because the axis starts at 95" is specific and constructive; "this chart is bad" is not.

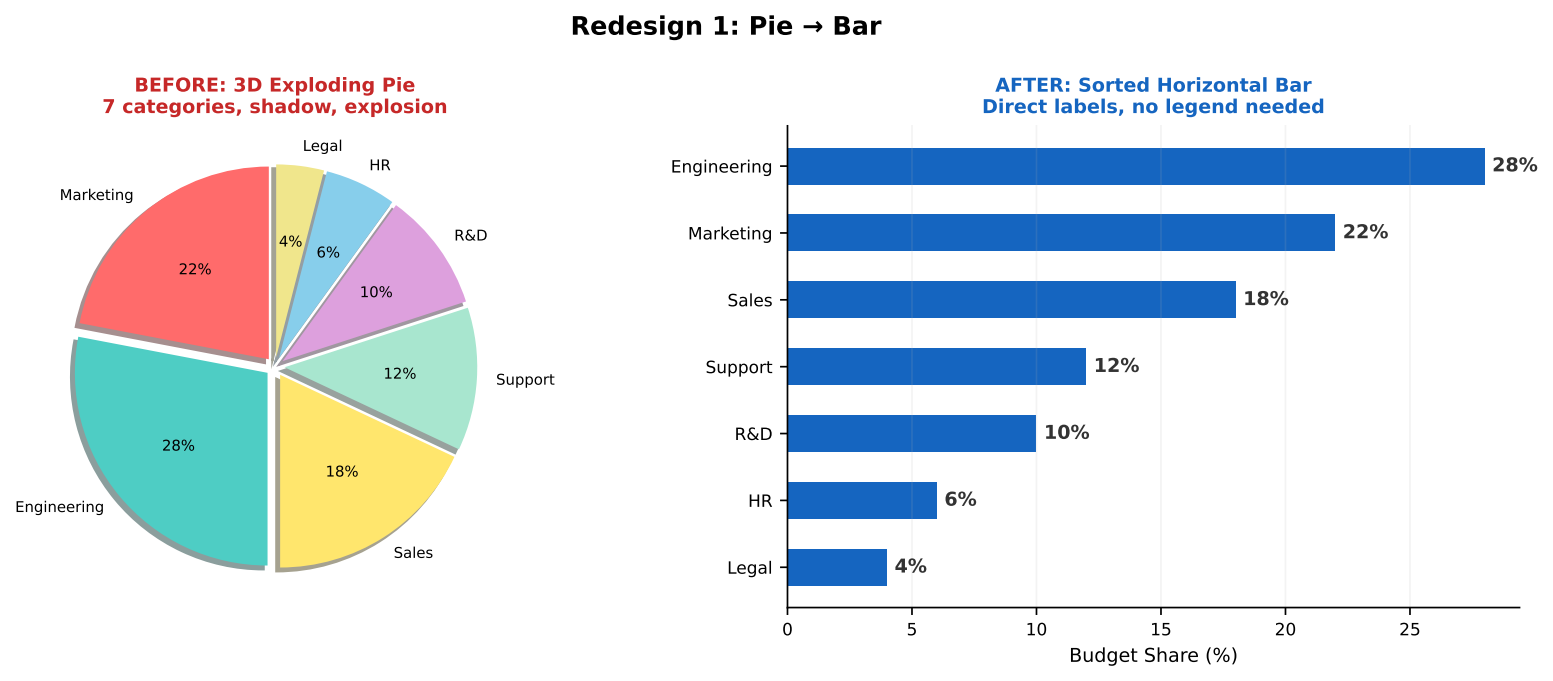
Five Redesign Case Studies

The Redesign Workflow



Every redesign follows five steps: identify the problem, diagnose which quadrant it belongs to, sketch an alternative, implement in code, and validate with a peer.

Redesign 1: 3-D Pie → Horizontal Bar



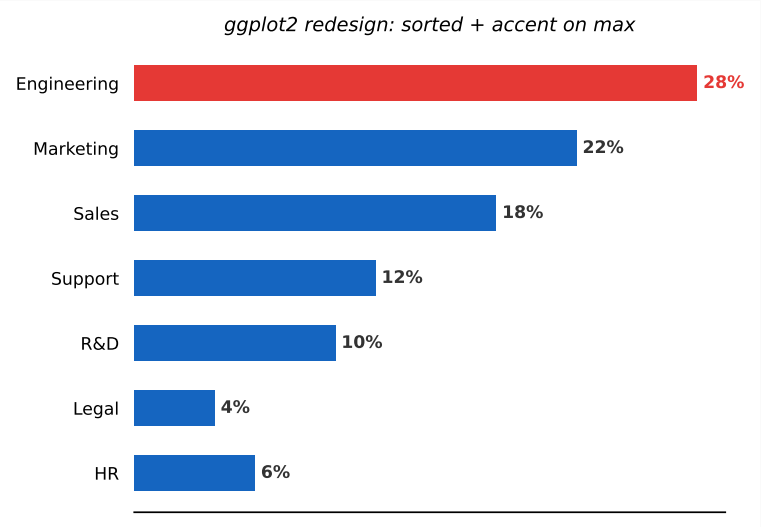
Data Insight

Diagnosis: Encoding (angle) + Craft (chartjunk: shadow, explosion).

Fix: Replace angle with position (horizontal bar), sort by value, add direct labels, remove all decoration.

Redesign 1 in R

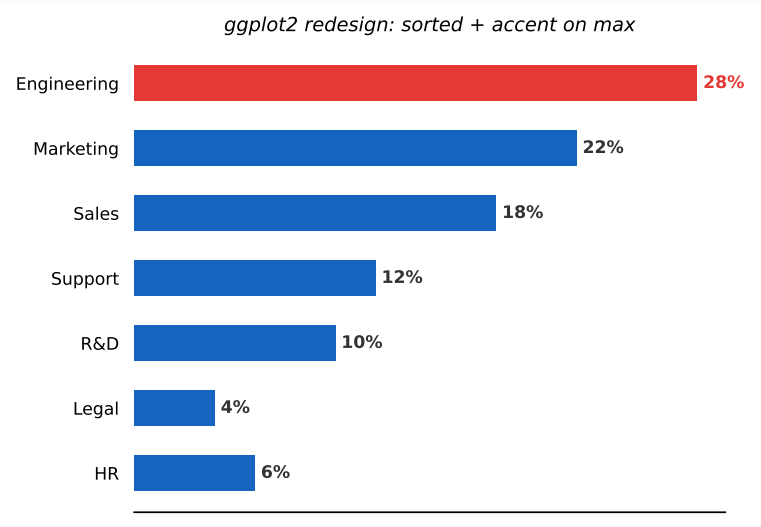
```r
library(tidyverse)

df <- tibble(
  dept = c("Marketing","Engineering
      ",
    "Sales","Support","R&D","HR",
    "Legal"),
  pct = c(22, 28, 18, 12, 10, 6, 4)
) |>
  mutate(dept = fct_reorder(dept,
      pct))

ggplot(df, aes(x = dept, y = pct))
    +
  geom_col(fill = "#1565C0",
           width = 0.55) +
  geom_text(aes(label = paste0(
    pct, "%")),
    hjust = -0.2, size = 3,
    fontface = "bold") +
  coord_flip() +
  theme_minimal() +
  theme(panel.grid.major.y =
    element_blank()) +
  labs(x = NULL,
    y = "Budget Share (%)",
```



Key moves: `fct_reorder()` sorts by value; `coord_flip()` makes bars horizontal; `geom_text()` adds direct labels.

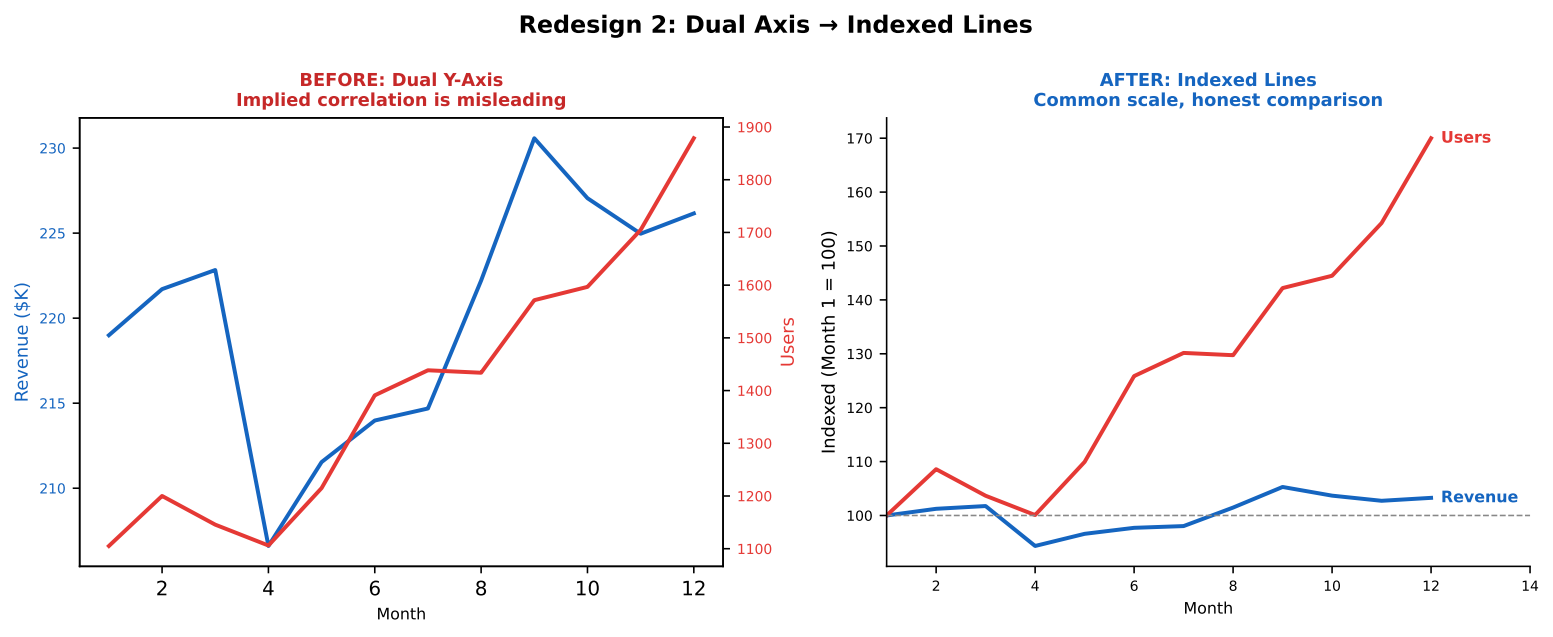
Redesign 1 in Python

```python
import matplotlib.pyplot as plt
import numpy as np

depts = ["HR","Legal","R&D",
  "Support","Sales","Marketing",
  "Engineering"]
pcts = [6, 4, 10, 12, 18, 22, 28]

# Sort ascending for barh
idx = np.argsort(pcts)
s_d = [depts[i] for i in idx]
s_p = [pcts[i] for i in idx]

fig, ax = plt.subplots()
colors = ["#1565C0"] * 7
colors[-1] = "#E53935"  # accent
ax.barh(s_d, s_p,
    color=colors, height=0.55)
for i, v in enumerate(s_p):
    ax.text(v + 0.3, i, f"{v}%",
        va="center", fontsize=9,
        fontweight="bold")

# Tufte cleanup
for sp in ["top","right","left"]:
```



Python equivalent: sort with `np.argsort()`, use `ax.barh()`, remove spines explicitly.

Redesign 2: Dual Y-Axis → Indexed Lines



Data Insight

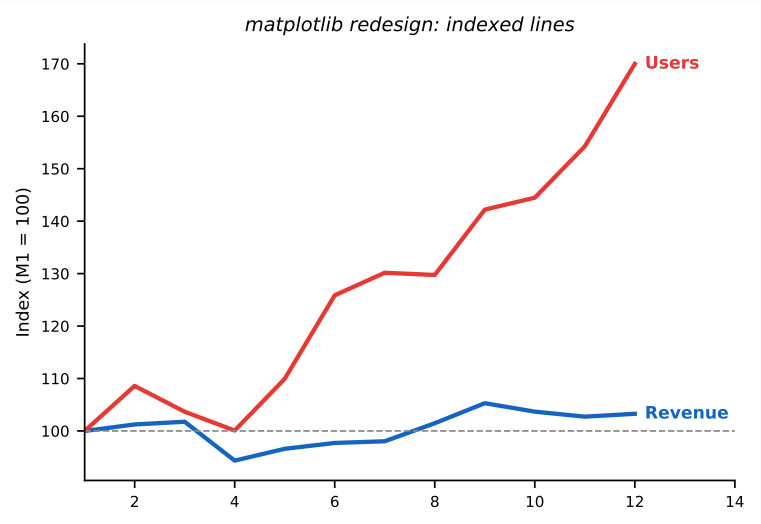
Diagnosis: Integrity (implied false correlation).

Fix: Index both series to 100 at month 1 so they share a common scale.

Relative change becomes directly comparable.

Redesign 2 in R

```r
# Index to 100 at first period
df_indexed <- df |>
  group_by(metric) |>
  mutate(
    index = value / first(value) *
        100
  )

ggplot(df_indexed,
  aes(x = month, y = index,
      color = metric)) +
  geom_line(linewidth = 1.2) +
  geom_hline(yintercept = 100,
    linetype = "dashed",
    color = "grey60") +

  # Direct labels at endpoint
  geom_text(
    data = df_indexed |>
      filter(month == max(month)),
    aes(label = metric),
    nudge_x = 0.5, size = 3,
    fontface = "bold") +
  theme_minimal() +
  theme(legend.position = "none") +
  labs(x = "Index (M1 = 100)")
```



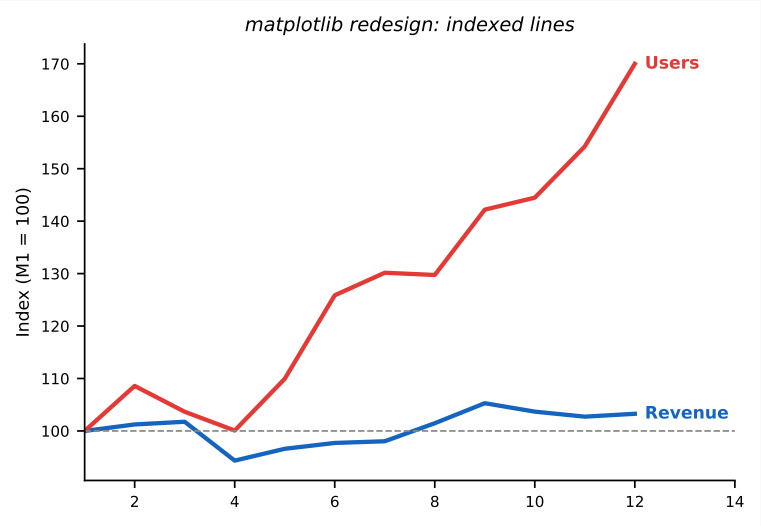
Indexing formula:

$\text{index}_t = \frac{v_t}{v_1} \times 100$

The reference line at 100 anchors "no change" visually.

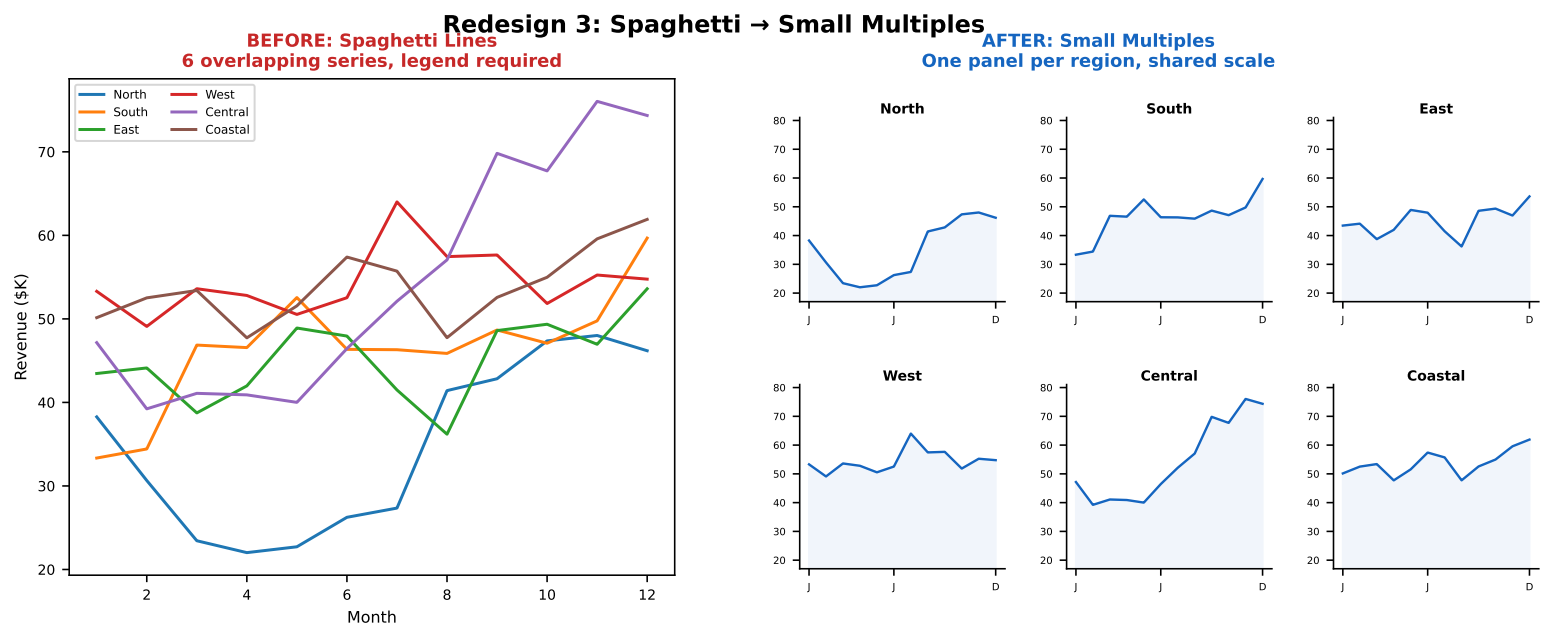
Redesign 2 in Python

```
# Index to 100 at month 1
rev_idx = revenue / revenue[0] *
    100
usr_idx = users / users[0] * 100

fig, ax = plt.subplots()
ax.plot(months, rev_idx,
    color="#1565C0", lw=2)
ax.plot(months, usr_idx,
    color="#E53935", lw=2)

# Reference line
ax.axhline(y=100, color="#888",
    lw=0.8, linestyle="--")

# Direct labels
ax.text(12.2, rev_idx[-1],
    "Revenue", fontsize=8,
    fontweight="bold",
    color="#1565C0", va="center")
ax.text(12.2, usr_idx[-1],
    "Users", fontsize=8,
    fontweight="bold",
    color="#E53935", va="center")

ax.set_xlim(1, 14)
```



NumPy vectorised indexing replaces the `group_by` + `mutate` pattern in R.

Redesign 3: Spaghetti Lines → Small Multiples

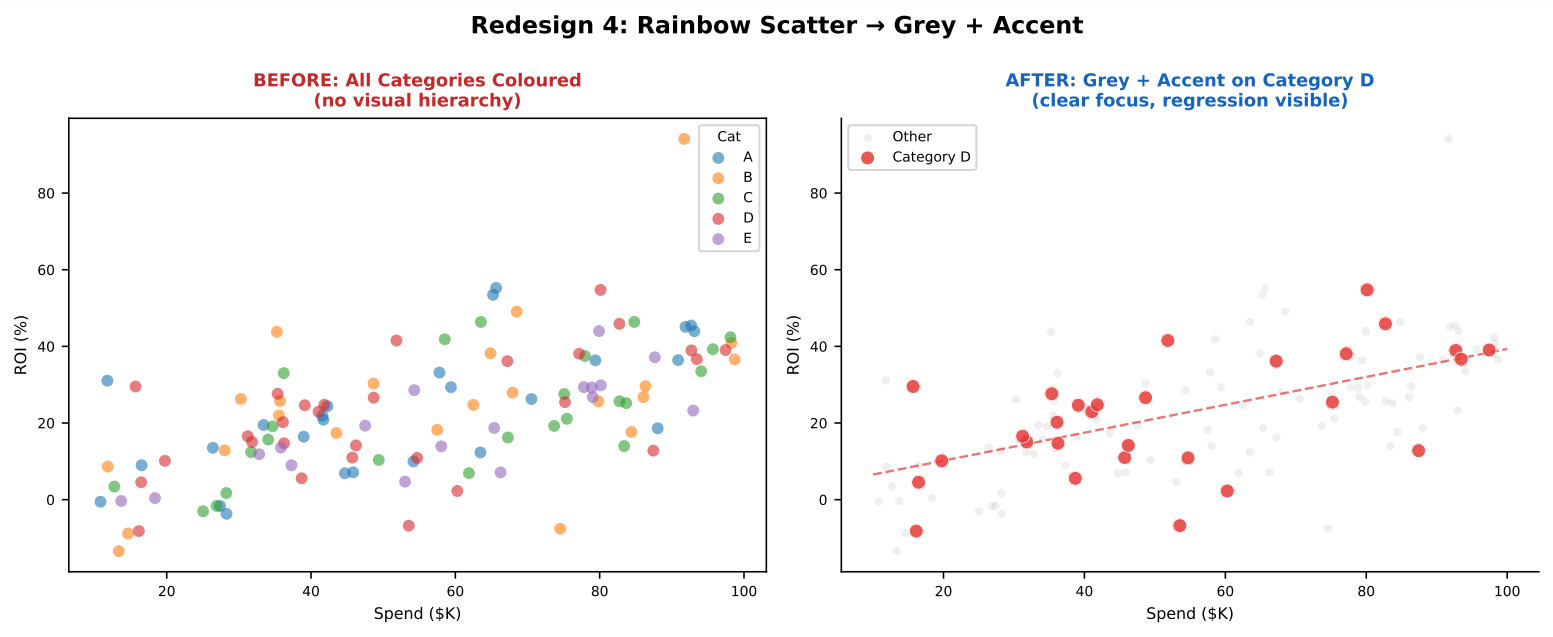


Data Insight

Diagnosis: Encoding (6 overlapping lines → conjunction search).

Fix: `facet_wrap()` / `subplots(sharey=True)` separates series by proximity (Gestalt). Shared y-scale ensures fair cross-panel comparison.

Redesign 4: Rainbow Scatter → Grey + Accent

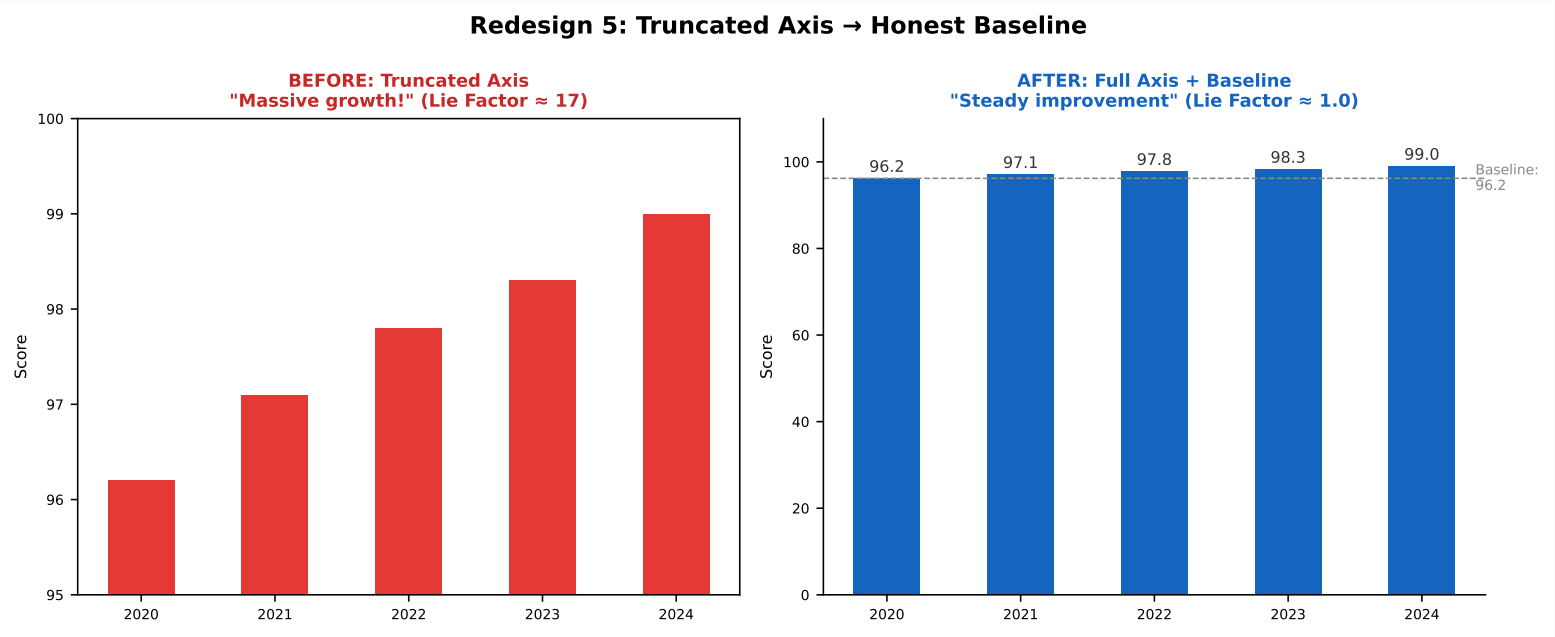


Data Insight

Diagnosis: Encoding (5 hues, no hierarchy) + Communication (no focal point).

Fix: Grey all non-focal points, accent the target category, add a regression line for the focus group only.

Redesign 5: Truncated Axis → Honest Baseline



Pro-Tip

Diagnosis: Integrity (lie factor ≈ 17).

Fix: Start y-axis at zero, add a dashed reference line at the first-period value, and label each bar directly. The visual impression shifts from "massive growth" to "steady improvement."

Peer Review Protocol

The Five-Step Peer Review

Peer Review Protocol: 5-Step Critique

1

First Impression (5 sec)

What do you see first?
What is the main message?

2

Data Integrity Check

Verify axes, scales, labels.
Calculate lie factor if suspicious.

3

Encoding Audit

Is each variable on the best channel?
Would a different chart work better?

4

Design Craft Review

Data-ink ratio? Chartjunk?
Typography? Whitespace?

5

Communication Test

Can a non-expert understand in 30 sec?
Is there a clear takeaway?

Giving Constructive Feedback

Structure (SBI Model):

- **S**ituation: "In the Q3 report chart. . ."
- **B**ehavior: ". . . the y-axis starts at 95, creating a lie factor of ∼17."
- **I**mpact: "This makes a 3% change look like a 50% change to leadership."

Always include:

- Which quadrant the issue falls in
- A specific, measurable diagnostic (lie factor, channel ranking)
- A concrete alternative ("try a zero-based axis with direct labels")

Pro-Tip
Lead with what works well before identifying problems. Frame improvements as "what if" alternatives, not "you should" commands.

In-Class Exercise: Critique & Redesign

Instructions (20 minutes):

1. Form pairs. Each person shows one chart from their own work or a published source.
2. Apply the **five-step peer review** protocol.
3. For each chart, identify:
   - The **strongest** quadrant (what works)
   - The **weakest** quadrant (what needs improvement)
4. Sketch a redesign on paper (3 minutes per chart).
5. Share one redesign with the class (2 minutes per pair).

Data Insight

The ability to critique constructively is the single most transferable skill from a visualization course. It applies to every chart you encounter for the rest of your career.

Synthesis

Five Redesign Patterns — Summary

Problem	Quadrant	Fix
3-D pie chart	Encoding + Craft	Horizontal bar
Dual y-axis	Integrity	Indexed lines
Spaghetti lines	Encoding	Small multiples
Rainbow scatter	Encoding + Comm.	Grey + accent
Truncated axis	Integrity	Zero baseline

Each pattern maps directly to a principle from Modules 1–6. The critique framework provides the diagnostic; the redesign implements the cure.

Key Takeaways

1. The **four-quadrant framework** (integrity, encoding, craft, communication) provides a systematic, non-subjective critique language.
2. Every redesign follows: **identify → diagnose → sketch → code → validate**.
3. Five high-frequency problems have canonical fixes: pie → bar, dual axis → index, spaghetti → facets, rainbow → grey-accent, truncated → zero-based.
4. **Peer review** using the SBI model (Situation, Behaviour, Impact) ensures feedback is specific and constructive.

Essential Readings for This Module

- Knaflic, C. N. (2015). *Storytelling with Data*, Chapter 3 (Clutter Is Your Enemy).
- Schwabish, J. (2021). *Better Data Visualizations*, Chapters 7–8 (Redesign Gallery).
- Wainer, H. (1984). "How to Display Data Badly." *The American Statistician*, 38(2), 137–147.
- Kirk, A. (2019). *Data Visualisation*, Chapter 9 (Interactivity and Annotation).

Next Module: R Environment Setup & Base Graphics (W01-M08)